NCFE Level 1 Functional Skills Qualification in Mathematics (603/5055/6)

Mark scheme: Practice P001268
v1.1

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Examiner Mark Scheme Guidance

Information

This guidance is intended to support NCFE examiners in the valid, reliable and consistent application of the relevant mark scheme version, against learner evidence generated during their external assessment.

This mark scheme provides:

- the total marks available for each question
- the subject content reference for each mark
- example process/methods and evidence of the types of responses expected for each mark
- (once confirmed) the pass mark for the relevant assessment version.

This mark scheme **must** be used for paper-based and online marking of the assessment version indicated**.**

Instructions and guidance on application

- All learners must receive the same treatment and should be marked fairly. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Learners must be rewarded for what they have shown they can do rather than penalised for things they have not done.
- Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Half marks must not be awarded.
- Examiners should be prepared to award zero marks if the learner's response is not worthy of credit according to the mark scheme.
- The mark scheme is a working document and may be added to at the standardisation to reflect valid alternative answers given by a learner.
- When in doubt regarding the application of the mark scheme to a learner's response, the Chief Examiner must be consulted.

This mark scheme provides the following information:

- section and activity information
- question number
- total marks available per question (top row, shaded) followed by
- attribution of individual marks per question
- problem solving (PS) and underpinning skill (UPS) attribution
- process/method or answers, as well as additional or alternative evidence; indicative of the subject content (SC) attribution
- any additional guidance, as required.

To support the valid, reliable and consistent marking of learner evidence, the following abbreviations are applied throughout the mark scheme:

Annotation	Explanation and use
FT	Follow through marks are applied when there are earlier mistakes in the method.
OE	Or equivalent marks are available for the justification of the answer being presented in a different form to the mark scheme i.e. 0.5 or ½.

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CAO	Correct answer only.
Their	'Their' refers to the learner's own values.
Seen	Seen refers to the requirement to see the stated value in the learner's response or working out.
Imp	Implied refers to the learner's response implying correct working out used but not seen.
Brackets	Indicates units are not required on final answers or for answers seen within working.
BOD	Benefit of doubt where learner handwriting may be difficult to interpret but previous working may indicate correct final answer.
Shaded	Indicates requirements for full marks to be awarded.

Version Control

Mark schemes are subject to version control. Examiners **must** ensure they have access to the latest version following each standardisation event.

Over time mark schemes will incorporate additional evidence captured and confirmed during standardisation events. Any additional evidence criteria will be captured in colour-coded text applicable to the dated standardisation event.

Note: Pass marks for Functional Skills external assessments are set in an awarding meeting, in which a combination of statistical analysis and professional judgement is used to determine the minimum required standard to achieve a pass in the assessment.

While different versions of the same assessment are designed to be of the same level of difficulty, variations in content can lead to the minimum required standard being represented by different marks across versions.

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Paper number: P001268				Version: 1.1	Pass mark: 38
(Section A) Activity 1: Wildflowers (Non-calculator Test)					
Q	**Marks**	**UPS / PS**	**Process and Answer**	**Additional or Alternative Evidence (with guidance)**	**SC**
1 (a)	**2**	**UPS**	1600	Award 2 marks if correct answer given	
	1		40 × 40		N6
	1		1600	CAO	N6
1 (b)	**2**	**UPS**	0.49 (m^2)	Award 2 marks if correct answer given	
	1		0.7 × 0.7		M22a
	1		0.49 (m^2)	CAO	N11b
1 (c)	**3**	**PS**	See below		
	1		0.5	FT their 0.49 rounded to one sig fig.	N12
	1		**their** 1600 ÷ their 0.5	FT their 1a & b	N15
	1		3200	FT their correct answer if using values from 1a & b	N11b
1 (d)	**5**	**PS**	8000	Award 5 marks if correct answer given	
	1		4 + 1 + 3 + 2 + 4 (+ 0) + 4 (+ 0) + 10 (+ 0 + 0) + 2 or 30	May be implied if 2.5 seen	H29a
	1		**their** 30 ÷ 12	May be implied if 2.5 seen	H29a
	1		2.5	CAO	H29a
	1		**their** 3200 × their 2.5	FT on **their** 1c	N5
	1		8000	CAO	N5
1 (e)	**1**	**UPS**	0.05	CAO	N16
1 (f)	**2**	**PS**	625 ($miles^2$)	Award 2 marks if correct answer given	
	1		$\frac{5}{100} \times 12500$ or 625	OE	N14
	1		625 ($miles^2$)	CAO	N14

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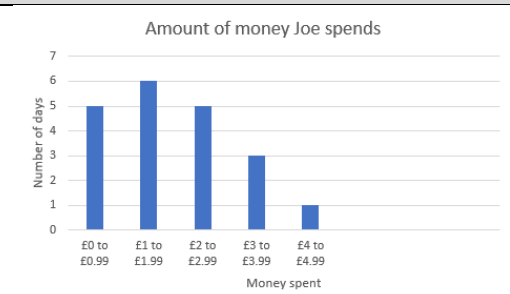
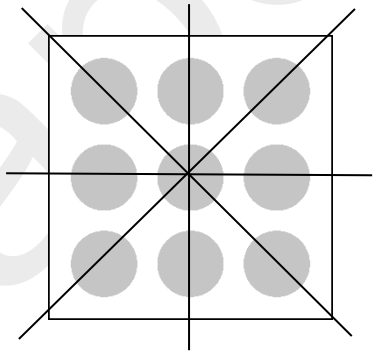
(Section B) Activity 2: Dinosaur cake (Calculator Test)					
Q	**Marks**	**UPS / PS**	**Process and Answer**	**Additional or Alternative Evidence (with guidance)**	**SC**
2 (a)	**2**	**UPS**	3200 (cm^3)	Award 2 marks if correct answer given	
	1		20 × 20 × 8 or 3200		M23
	1		3200 (cm^3)	CAO	M23
2 (b)	**4**	**PS**	165 (g)	Award 4 marks if correct answer given	
	1		2.4 × 1000 or 3200 ÷ 1000		M20c
	1		their (2400) ÷ 3200 or 0.75	OE	N17
	1		their 0.75 × 220 or (3 × 220) ÷ 4	OE Award 2 middle marks if 220 / 3200 × 2400 = (165) or 220 / 4 × 3 = (165) seen	N17
	1		165 (g)	CAO	N17
2 (c)	**2**	**PS**	See below		
	1		0.6 and 0.66… seen or $\frac{9}{15}$ and $\frac{10}{15}$ seen or 60% and 66-67%		N8b
	1		Tin A with valid comparable working seen	dependent on 1st mark being awarded	N8b
2 (d)	**2**	**PS**	See below		
	1		$1\frac{1}{4}$ or 1.25	OE Accept 125%	N8b
	1		Yes and $\frac{1}{4} = 25\%$ or 0.25 = 25%	Or implied from correct statement, ie $1\frac{1}{4}$ = 1.25	N16
2 (e)	**3**	**PS**	See below		
	1		6 × 90 or 6 × 0.9 or 540 or 5.4 or £5.40		N17
	1		4 × 120 or 4 × 1.2 or 480 or 4.8 or £4.80		N17
	1		Toppers supported by working	Comparison must between two values in the same units, from price × packs	N17

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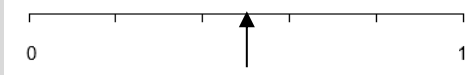
2 (f)	**2**	**PS**	See below		
	1		$\frac{6}{24}$	OE	H31
	1		0.25	FT their $\frac{6}{24}$	N16

Activity 3: Student finance (Calculator)					
Q	**Marks**	**UPS / PS**	**Process and Answer**	**Additional or Alternative Evidence (with guidance)**	**SC**
3 (a)	**4**	**PS**	See below		
	Alternative method 1				
	1		27120 – 25000 or 2120		N1b
	1		their 2120 × 15 ÷ 100 or 212 + 106 or 0.15 × 2120 or 318	OE method for finding 15%	N14
	1		318 ÷ 12		N17
	1		(£)26.50	CAO to 2 dp	N17
	Alternative method 2				
	1		27120 ÷ 12 and 25000 ÷ 12 or 2260 and 2083.3…		N17
	1		their 2260 – their 2083.3… or 176.6…..		N1b
	1		their 176.6… × 0.15	OE method for finding 15%	N14
	1		(£)26.50	Allow (£)26.49 to 2dp Allow 2650p or 2649p	N17
3 (b)	**3**	**UPS**	(£)1237.50	Award 3 marks if correct answer given	
	Alternative method 1				
	1		$\frac{5}{100}$×1125 or 56.25	OE May be implied if 112.5 seen	M18
	1		their 56.25 × 2 or 112.5		M18
	1		(£)1237.50	CAO to 2dp	M18
	Alternative method 2				
	1		$\frac{1125}{10}\times\frac{1}{2}$ or 56.25	OE	M18

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	1		their 56.25 × 2 or 112.5		M18
	1		(£)1237.50	CAO to 2 dp	M18
3 (c)	**3**	**PS**	See below		
	1		Correct frequencies 5, 3 and 1		H28a
	1		At least 2 bars correct for their freqs		H28b
	1		All bars correct heights for their freqs		H28b
3 (d)	**2**	**UPS**	See below		
	1		1 for both horizontal and vertical line correct Do not award if additional lines are drawn ie maximum of 2 lines		M24b
	1		1 for both diagonals lines correct Do not award if additional lines drawn		M24b
3 (e)	**1**	**PS**	Open line because (0.4) is biggest number/probability		H30b
3 (f)	**2**	**PS**	See below		
	1		Open Line (£)6.91	CAO	H29b
	1		The Abbey Trust (£)9.23	FT 35.56 – their	H27a

Activity 4:	**Student services**		**(Calculator)**		
Q	**Marks**	**UPS / PS**	**Process and Answer**	**Additional or Alternative Evidence (with guidance)**	**SC**
4 (a)	**2**	**PS**	21(.0) and 9.9	Award 2 marks if correct answer given	
	1		21(.0)		M25b
	1		9.9		N9
4 (b)	**3**	**PS**	1247.4 (cm^2)	Award 3 marks if correct answer given	
	Alternative method 1				

	1		29.7 × 21.0 or 623.7		M22a
	1		their 632.7 × 2		M25b
	1		1247.4 (cm^2)	allow 1247 but not 1248	M22a
	Alternative method 2				
	1		their 21.0 × their 9.9 or 207.9		M22a
	1		their 207.9 × 6		M25b
	1		1247.4 (cm^2)	allow 1247 but not 1248	M22a
4 (c)	**1**	**PS**	No and dimensions should be 40 and 50 or 90 should not be on plan		M25a
4 (d)	**1**	**UPS**	Any hexagon drawn		M24a
4 (e)	**1**	**UPS**	120° ±(2 degrees)		M26b
4 (f)	**1**	**UPS**	Obtuse	CAO	M24c
4 (g)	**3**	**PS**	See below		
	1		$\frac{50}{100}\times 25000$ or 12500	OE correct method	N14
	1		$\frac{40}{100}\times 32000$ or 12800	OE correct method	N14
	1		This year	Must be both correct to award all 3 marks	N13
4 (h)	**2**	**PS**	See below		
	1		Scale should go up in 10s	Any comment that relates to scale not being consistent. Do not allow 'scale is wrong'	H27c
	1		Well-being bar the wrong height		H27c
4 (i)	**1**	**UPS**		Accept point in middle; mark intention	H30a

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